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Classification of urban form of Jaffna

The urban form of a town refers to the physical layout, arrangement, and structure of its built environment. It encompasses the spatial organization of buildings, streets, public spaces, and other elements that shape the overall character and appearance of the town. The urban form includes factors such as building density, architectural styles, land use patterns, transportation infrastructure, open spaces, and the relationship between different parts of the town. It influences how people navigate and interact within the town, as well as the functionality, aesthetics, and livability of the urban environment. The urban form of a town can vary widely depending on historical, cultural, economic, and planning factors.

According to these factors, I choose to classify the urban form of Jaffna town based on its physical characteristics and the functionalities.



Figure 2 Satellite map of Jaffna taken from google earth.

The town's narrow streets create a unique atmosphere, while vibrant marketplaces like the Jaffna Vegetable Market showcase the local economy. Jaffna town area include its rich cultural heritage reflected in its historic architecture, such as the Jaffna Fort and traditional Tamil-style houses. Jaffna's numerous religious sites, including Nallur Kovil and Jaffna Cathedral, highlight the town's spiritual significance. The presence of landmarks like Jaffna Fort and religious sites, the layout of major roads and intersections, the distribution of open spaces and parks can be observed from the satellite map of Jaffna.

Several factors can be observed in the residential and commercial building patterns of Jaffna town. By analyzing the building characteristics of Jaffna city, helps to identify the urban form of its.

- The buildings in the central area of Jaffna town, particularly the commercial district, are densely clustered and concentrated. This concentration reflects the focus of economic and commercial activities around the central core.
- The buildings in the central area often exhibit a mix of residential and commercial uses. It is common to find shops, offices, and other commercial establishments on the ground floor of buildings, with residential units on the upper floors. This mixed-use development pattern contributes to the vibrancy and vitality of the central area.
- In the central core of Jaffna town, taller buildings are more prevalent compared to the surrounding areas. This is often due to the higher land values and the demand for commercial and office spaces, which encourage vertical development.
- Buildings in the Jaffna central area typically have direct frontage along the main streets, contributing to the walkability and accessibility of the commercial district. The street-facing facades of buildings often feature storefronts, entrances, and displays to attract customers and engage pedestrians.
- The development density of Jaffna town is generally higher in the central area, with buildings closely spaced and occupying most available land. This density reflects the efficient use of limited land resources and the desire to maximize the utilization of prime locations.

The below map shows the road network and building patterns of Jaffna city. Dark yellow center part of the town area is the CBD of Jaffna town. Main offices, public bus stand, general hospital, and commercial centers are concentrated in that core area. Other commercial centers and the business units are located along the main road which are towards to other sub-urbans. Light yellow which is cover the biggest part of the town indicates the residential buildings in the Jaffna region.



Figure 3 Satellite map of building distribution in Jaffna town

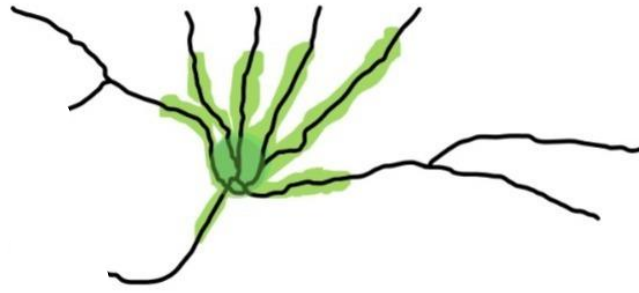


Figure 4 Commercial activity pattern and road network of Jaffna town

Above map shows the central core of the Jaffna town and the density of buildings in the town and towards the Jaffna town area.

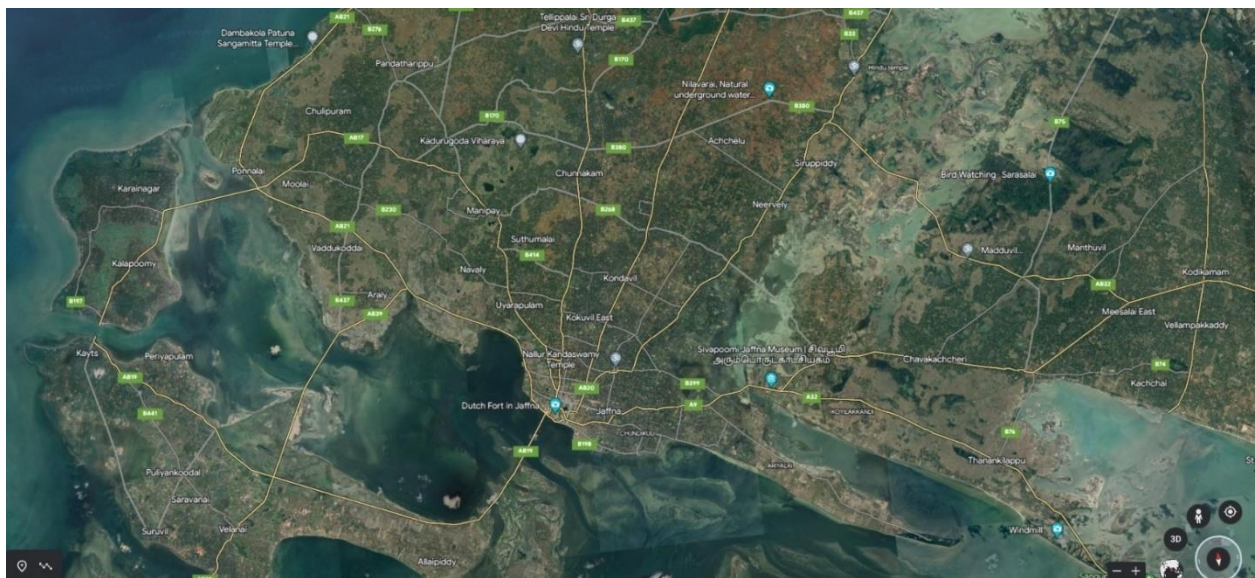


Figure 5 Satellite map of Road network of Jaffna taken from google.

The road network of Jaffna town is primarily organized around a central core, which is the main center of activity and serves as the hub for transportation and commercial activities such as Jaffna bus stand, Jaffna general hospital, Jaffna public market area. The road network radiates outward from this central area in a radial pattern, with roads extending from the center to the periphery of the Jaffna town. These radial roads provide connectivity to other sub-urbans of the town and facilitate access to and from the central core.

The roads leading to the central area are often wider and more prominent, accommodating higher volumes of traffic. As one moves away from the Jaffna center, the roads generally become narrower and quieter, catering more to local traffic and serving residential areas. The road network in the central area is typically more densely interconnected, with multiple roads and intersections facilitating easy movement within the core. This pattern enhances accessibility, facilitates movement, and contributes to the overall functionality and organization of the Jaffna town.

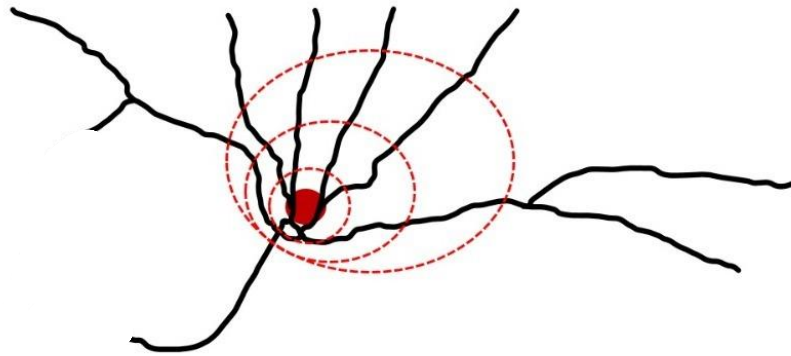


Figure 6 Monocentric urban form of Jaffna town

The monocentric urban form of Jaffna town is clearly confirmed by the above classification of building forms and road network of Jaffna town. The presence of a central business district with a high concentration of commercial and office buildings indicates the existence of a core area in the town. The road network radiates outward from this central area, with major roads connecting to surrounding neighborhoods and suburbs. The building patterns show a denser development near the town center, gradually transitioning into lower density as one moves away. These factors collectively suggest the presence of a monocentric urban form in Jaffna town.

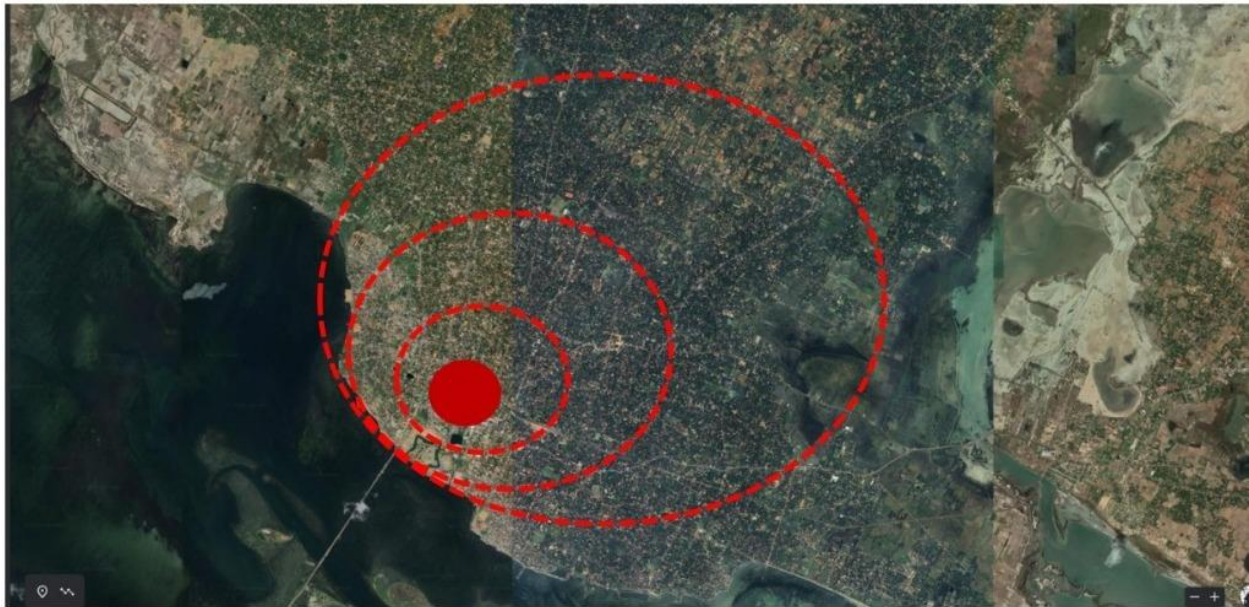


Figure 7 Monocentric urban form Satellite map of the Jaffna town

Here are some of the positive qualities commonly found in Jaffna town,

1. Rich cultural heritage and historical significance: Jaffna town is known for its deep cultural roots and historical landmarks, such as the Jaffna Fort and Nallur Kandaswamy Temple. These attractions draw tourists and contribute to the town's unique identity.
2. Vibrant local markets and commercial activities: Jaffna town has bustling local markets like the Jaffna Central Market, where residents can find a wide range of fresh produce, spices, handicrafts, and traditional items. These markets create a lively atmosphere and support local businesses.
3. Close-knit community and strong social ties: Jaffna town has a close-knit community where people maintain strong social bonds and communal values. This sense of community fosters a supportive and inclusive environment.
4. Availability of educational institutions: Jaffna town is home to various educational institutions, including schools, colleges, and universities. These institutions provide opportunities for learning and intellectual growth, contributing to the town's educational development.
5. Well-preserved traditional architecture: The town showcases well-preserved traditional architecture, including colonial-era buildings and traditional Jaffna-style houses. This architectural heritage adds to the town's aesthetic appeal and cultural value.

Here are some of the positive qualities commonly found in Jaffna town,

1. Limited infrastructure and outdated facilities: Jaffna town faces challenges in terms of inadequate infrastructure, including road networks, public transportation, and utility services. The existing facilities may also require upgrading to meet modern standards.
2. Traffic congestion and inadequate transportation options: The town's road network struggles to handle increasing traffic demands, resulting in congestion during peak hours. Limited public transportation options further exacerbate the issue.
3. Insufficient public amenities and services: Jaffna town may lack sufficient public amenities such as parks, recreational facilities, healthcare centers, and community spaces. Residents may face challenges in accessing essential services.
4. Impact of past conflict and post-war challenges: Jaffna town has faced the consequences of a prolonged civil war, including infrastructure damage and psychological trauma. The process of post-war recovery and reconciliation is an ongoing challenge for the town.
5. Lack of job opportunities and economic diversification: The town's economy heavily relies on a few sectors, such as agriculture and small-scale industries. The limited availability of diverse job opportunities may result in unemployment and underemployment.
6. Environmental degradation and pollution: Rapid urbanization and inadequate waste management practices can lead to environmental degradation and pollution. Ensuring sustainable development and environmental protection is crucial for the town's long-term well-being.
7. Limited recreational and leisure options: Jaffna town may have limited recreational facilities and leisure activities for residents and visitors. Developing more parks, sports facilities, and cultural spaces can enhance the quality of life in the town.

Many of the qualities of the town directly resulted in its urban form,

- Rich cultural heritage and historical significance
old Jaffna kingdom town was Nallur. Current Jaffna town is locating very close to Nallur, and the Jaffna town is very close to Jaffna Dutch fort. These historic and cultural factors impact on the Jaffna town formation.
- Vibrant local markets and commercial activities
Commercial activities and market encourage the activity pattern in the central core of the town area. Other sub-urbans activities also directed towards to Jaffna town.
- Availability of educational institutions
Main schools, Private tutorials, and Jaffna University is locating very close to Jaffna town. these characteristics also resulted directly to its monocentric urban form.
- Employment opportunities
All these commercial activities and businesses locating in the town center encourage the job opportunities and encourage the movement of the public towards the town.